

Learning Emergent Modular Representations in Multi-modality Medical Vision Foundation Models (*Supplementary Materials*)

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A Theoretical Foundation

A.1 Theoretical Modeling of the Non-IID Problem

Multi-modality MVFM pretraining violates the IID assumption [10, 63] due to heterogeneous imaging physics, protocols, and contexts [72, 79, 81]. This section provides a formal characterization of the Non-IID setting, its effect on self-supervised learning, and the resulting modality-dominant representation collapse.

A.1.1 Preliminary 1: Multi-Modality Distributions. Let $\mathcal{D} = \{1, \dots, D\}$ denote the set of imaging modalities, where each modality $d \in \mathcal{D}$ induces a distribution P_d over input space $\mathcal{X}$. Uni-modality MVFM pretraining assumes IID samples $x_i \sim P$ drawn from a single distribution P . Multi-modality pretraining samples from the mixture

$$P_{\text{mix}} = \sum_{d=1}^D \rho_d P_d, \rho_d \geq 0, \sum_d \rho_d = 1, \quad (1)$$

where ρ_d is the sampling proportion of modality d . Each sample is effectively drawn as

$$d_i \sim \rho, \quad x_i \sim P_{d_i}, \quad (2)$$

which violates the standard IID assumption that all $x_i \sim P$ for a single underlying P . This induces distribution shift across modalities, resulting in the *Non-IID setting* that is characterized by pronounced inter-modality discrepancies and disjoint modality clusters in feature space [5].

Definition A.1 (Non-IID Multi-Modality Setting). A dataset is *Non-IID* if its samples are drawn from a mixture of distributions $\{P_d\}_{d=1}^D$ satisfying $\exists d \neq d', P_d \neq P_{d'}$.

A.1.2 Preliminary 2: Self-supervised Learning in Non-IID setting. Let $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^d$ be a feature extractor, and let $\mathcal{T}$ denote the self-supervised pretext transformations. A self-supervised objective is formulated as

$$\begin{aligned} \mathcal{L}_{\text{self}} &= \mathbb{E}_{x \sim P_{\text{mix}}} [\ell(f_\theta(x), \mathcal{T}(x))] \\ &= \sum_{d=1}^D \rho_d \mathbb{E}_{x \sim P_d} [\ell(f_\theta(x), \mathcal{T}(x))] \end{aligned} \quad (3)$$

where the loss ℓ induces a pretext task in the self-supervised learning (SSL). Standard SSL assumes IID sampling from a single P [63], which is violated under Non-IID sampling.

Definition A.2 (Feature Decomposition). We decompose the learned representation $f_\theta(x)$ as an arbitrary layer as a non-linear combination of a modality-invariant semantic component and a modality-specific component:

$$f_\theta(x) = g_\theta^{\text{sem}}(x) \oplus g_\theta^{\text{mod}}(x, d), \quad (4)$$

where $g_\theta^{\text{sem}}(x)$ captures modality-invariant semantics, and $g_\theta^{\text{mod}}(x, d)$ captures modality-specific biases, and $\oplus$ denotes integration.

A.1.3 Representation Collapse as Degenerate Optimum.

THEOREM A.3 (SEMANTIC ATTENUATION AND MODALITY COLLAPSE UNDER NON-IID SSL). Let $\Delta_{\text{mod}}(\mathcal{F})$ denote the intrinsic distribution discrepancy across modalities in the hypothesis class $\mathcal{F}$. Without explicit control, i.e., optimizing $\mathcal{L}_{\text{self}}$ only, as the modality discrepancy $\Delta_{\text{mod}}(\mathcal{F})$ increases, the optimizer tends to prioritize modality-specific features. The relative magnitudes of the decomposed features will be:

$$\frac{\|g_{\theta^*}^{\text{mod}}\|}{\|g_{\theta^*}^{\text{sem}}\|} \rightarrow \mathcal{O}(\Delta_{\text{mod}}(\mathcal{F})) \quad (5)$$

This semantic attenuation means that the network over-relies on $\mathcal{T}(x)$'s modality-dependent statistics for quick loss minimization, leading to representation collapse where effective generalization across modalities is lost.

PROOF SKETCH. In the Non-IID setting, minimizing the mixed objective $\mathcal{L}_{\text{self}}$ (Equ.(3)) is easier by finding shortcuts that reduce local modality risks $\mathbb{E}_{x \sim P_d} [\dots]$ independently. The modality-specific component g_θ^{mod} offers a high-variance, low-effort path to minimize local $\mathcal{L}_{\text{self}}$ terms. In contrast, the modality-invariant component g_θ^{sem} must satisfy a consistent objective across all highly heterogeneous P_d , requiring significantly more optimization effort. This difference in gradient efficiency causes g_θ^{mod} to be amplified and g_θ^{sem} to be suppressed, leading to the collapsed representation dominated by modality biases. $\square$

A.2 Modular Networks for Non-IID Problem

Our Director-Experts (DEX) modular network, is designed to explicitly regulate the feature decomposition $f_\theta(x) = g_\theta^{\text{sem}}(x) \oplus g_\theta^{\text{mod}}(x, d)$ during optimization, thus counteracting semantic attenuation.

Definition A.4 (Director-Experts (DEX)). Our DEX models a network F as a hierarchical modular network $F = M_L \circ M_{L-1} \circ \dots \circ M_1$, where for an input f_{l-1} , each module M_l ($l = 1, \dots, L$) performs:

- (1) **Specialization** Φ_{spec}^l : Multiple expert modules dynamically partition the parameter space to learn modality-specific features $\hat{f}_l$.

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- (2) **Coordination** Φ_{co}^l : A director module coordinates experts' knowledge by aligning their output $\hat{f}_l$ to a shared latent space $\mathcal{D}(f_{l-1})$

Excluding the other auxiliary losses, DEX is optimized by a self-supervised learning loss and an alignment loss:

$$\mathcal{L}_{DEX} = \mathcal{L}_{self} + \frac{\lambda_{co}}{L} \sum_{l=1}^L \alpha^l \mathcal{L}_{co}^l, \quad (6)$$

where λ_{align} is the alignment weight and $\alpha^l = \frac{1}{L-(l-1)}$ is the layer-wise factor.

A.2.1 The Regularization Effect of Alignment Loss $\mathcal{L}_{align}$. The director module $\mathcal{D}$ maintains a representation $\hat{f}^D$ that serves as an aggregation of multi-modality knowledge via the Group Exponential Moving Average (GEMA) $\eta^*(\theta) = \mathcal{G}_{EMA}(\theta, \eta)$. $\hat{f}^D$ thus models the stable, shared (semantic) component $g_\theta^{sem}(x)$ across modalities. The alignment loss, based on cosine similarity, actively forces the expert's output $\hat{f}^E$ towards $\hat{f}^D$:

$$\mathcal{L}_{align}(\hat{f}^E, \hat{f}^D) = 1 - \frac{\hat{f}^E \hat{f}^D}{\|\hat{f}^E\| \|\hat{f}^D\|} \quad (7)$$

LEMMA A.5 (DEX IMPOSES UPPER BOUNDS ON MODALITY RESIDUALS). *The alignment loss $\mathcal{L}_{co}$ effectively penalizes the modality-specific residual g_θ^{mod} . Specifically, the expected norm of the modality residual at layer l , $\mathbb{E}[\|g_l^{mod}\|]$, is bounded by:*

$$\mathbb{E}[\|g_l^{mod}\|] \leq K_l = \mathcal{O}\left(\frac{L_{self}}{\lambda_{co} \alpha^l c_l}\right), \quad (8)$$

where L_{self} is the Lipschitz constant of $\mathcal{L}_{self}$, $c_l > 0$ is a constant related to the alignment loss property.

The factor α^l ensures that deeper layers have a greater penalty on modality residuals, enforcing progressive contraction towards a common representation, consistent with representation evolution.

A.2.2 Mitigation of Semantic Attenuation.

THEOREM A.6 (DEX MITIGATES SEMANTIC ATTENUATION UNDER NON-IID SSL). *By setting the $\mathcal{L}_{co}$ weights $\lambda_{co} \alpha^l$ sufficiently large in deeper layers, DEX effectively contracts the modality discrepancy, reducing the overall effective discrepancy $\Delta_{mod}^{DEX}(\mathcal{F})$ compared to the original $\Delta_{mod}(\mathcal{F})$. As a result, any minimizer f_θ^* of $\mathcal{L}_{DEX}$ satisfies:*

$$\|g_{\theta^*}^{sem}\| \geq C_{sem} \quad \text{and} \quad \|g_{\theta^*}^{mod}\| \leq C_{mod} \quad (9)$$

where $C_{sem} > 0$ is a lower bound independent of $\Delta_{mod}(\mathcal{F})$, and C_{mod} is an upper bound. IAR thus simultaneously prevents semantic attenuation by requiring the Director to maintain a strong shared latent space, and prevents modality-dominated collapse by bounding the residual component.

PROOF SKETCH. The expert modules use the self-supervised learning loss $\mathcal{L}_{self}$ to learn high-variance, modality-specific features (g_θ^{mod}) while the Director, updated via the GEMA, learns the low-variance, modality-invariant features (g_θ^{sem}). The $\mathcal{L}_{align}$ loss then acts as an explicit ℓ_2 -norm-like penalty on the distance between $\hat{f}^A$ and $\hat{f}^D$, effectively regularizing the magnitude of the g_θ^{mod} (Lemma A.5), thus balancing the objective and ensuring that the representation is dominated by semantics, not modality-specific biases. $\square$

B Experiment Details

B.1 Datasets

B.1.1 Pretraining dataset: Medical Vision Universe (MedVerse). Our curated MedVerse dataset is a large-scale multi-modality MV dataset collected from publicly available medical imaging datasets and aggregated to support training general-purpose MVFMs (Tab.1). It contains more than 4.97 million medical images and slices across 10 modalities sourced from more than 40 well-established medical datasets, covering X-ray, CT, PET, MR, ultrasound, fundus, OCT, pathology, endoscopy, and clinical photography. It covers a broad spectrum of anatomical regions such as the chest, breast, bone, heart, brain, abdomen, eye, fetus, prostate, and skin. To improve the overall data integrity, we remove low-quality, corrupted, or abnormally small images, as well as highly repetitive or near-duplicate samples. This results in a cleaner, more diverse, and more representative corpus for pretraining.

1) CT (821K): The CT component of MedVerse comprises whole-body, cardiac, and brain CT datasets, including AutoPET-CT [20], ImageCAS [74], and TopBrain-CT [71], totaling more than 821K slices. To standardize voxel intensities across heterogeneous scanners and protocols, all CT volumes are intensity-normalized by clipping the Hounsfield Unit (HU) range to $[-200, 400]$, which covers the majority of soft-tissue and organ structures relevant to downstream tasks. Following clipping, intensities are linearly rescaled to $[0, 1]$ to ensure compatibility with other modalities in MedVerse. To further improve data quality, we discard slices with minimal anatomical content, defined as those with a foreground (non-air) pixel ratio below 0.05. This filtering step removes blank, near-empty, or scanner-padding slices commonly found at volume boundaries.

2) PET (706K): The PET portion of MedVerse from the AutoPET [20] dataset comprising more than 706K slices. It includes both raw PET intensity volumes and standardized uptake value (SUV) volumes, enabling models to learn representations invariant to acquisition differences and thus improving cross-domain generalization. For raw PET data, we apply a three-stage normalization pipeline: (1) intensities are first clipped using the 1% and 99% percentile values to suppress extreme noise and scanner artifacts; (2) the clipped volumes are then standardized using z-score normalization; (3) the standardized values are finally mapped to $[0, 1]$ via min-max normalization. For SUV volumes, we follow clinical conventions by clipping values to the range $[0, 20]$, which captures typical physiological and pathological uptake levels, and subsequently linearly normalizing all values to $[0, 1]$. To ensure anatomical relevance and remove non-informative slices from both modalities, we exclude slices in which the foreground ratio (non-air uptake) is below 0.01.

3) MR (1.75M): The MRI component of MedVerse consists of whole-body, cardiac, breast, abdominal, and brain MRI datasets, including Totalsegmenter-MR [2], ACDC [6], MRNet [8], Duke-Breast-Cancer-MRI [60], Prostate-MRI-US-Biopsy [19, 50], TopBrain-MR [71], LLDMMR [43], and BraTS-GLI [14], comprising more than 1.75M slices. To standardize voxel intensities across heterogeneous sources, all MRI volumes undergo intensity normalization by clipping each volume's intensity distribution using the 1% and 99% percentile values, which suppresses sequence-specific outliers and scanner-dependent artifacts. Following clipping, voxel intensities

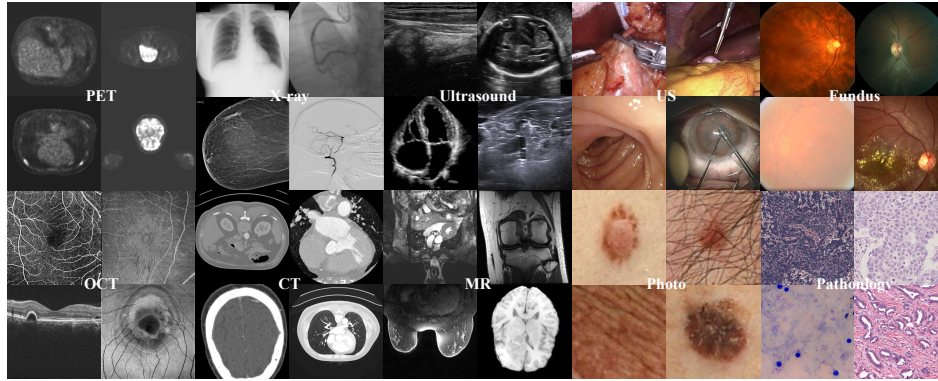


Figure 1: Examples in our MedVerse dataset. It has 10 modalities with diverse patterns and clinical targets, providing a large amount of medical image knowledge for FMs and enabling them to generalize to different tasks.

are linearly rescaled to $[0, 1]$, facilitating consistent numerical ranges across different MRI contrasts and ensuring compatibility with other modalities in MedVerse. We discard slices with minimal anatomical content, defined as those with a foreground (non-air/tissue) pixel ratio below 0.05, thereby removing blank or non-informative slices typically appearing at volume boundaries.

4) X-ray (468K): The X-ray component of MedVerse consists of a diverse collection of large-scale public datasets, including ChestX-ray8 [68], CheXpert [30], Mammo-Bench [7], MURA [59], vrad2 [58], ADSD [12], ARCADE [56], and XACD [45], totaling more than 468K radiographs. These datasets span a wide range of anatomical regions, such as the chest, breast, bone, tooth, and coronary artery, and include both diagnostic and screening scenarios. All images are normalized by dividing pixel intensities by 255, ensuring consistent numerical ranges across sources.

5) Ultrasound (68K): The ultrasound component of MedVerse covers multiple anatomical regions, including cardiac, breast, thyroid, fetal, and peripheral nerve images, and aggregates both static images and cine sequences from public datasets such as EchoNet-Dynamic [52, 53] and related ultrasound benchmarks [3, 16, 62, 65], thus comprising more than 68K images. To make video-based data compatible with our 2D pretraining pipeline, we subsample ultrasound videos by storing one frame every 30 frames, which preserves the temporal diversity of cardiac motion while controlling redundancy and storage cost. All ultrasound images and sampled video frames are normalized by dividing pixel intensities by 255, resulting in values in the range $[0, 1]$ and ensuring numerical consistency across datasets and acquisition devices.

6) Fundus (96K): The fundus component of MedVerse includes a diverse set of ophthalmology datasets such as Refuge2 [18, 51], MESSIDOR [15], APTOS2019 [29], and Eyepacs [22], comprising more than 96K images. These datasets provide high-resolution color fundus photographs captured under varying illumination conditions, camera systems, and clinical screening settings. To ensure consistent numerical ranges across heterogeneous sources, all fundus images are normalized by dividing RGB pixel intensities by 255, mapping them to the $[0, 1]$ range while preserving color fidelity.

7) OCT (110K): The OCT component of MedVerse consists of retinal OCT datasets, including OCTA500 [39], ROSE [44], and other public OCT benchmarks [34], comprising more than 110K images. These datasets provide cross-sectional scans acquired under varying scanning protocols, resolutions, and device manufacturers. To standardize appearance across sources, all OCT images are normalized by dividing pixel intensities by 255, mapping values to the $[0, 1]$ range while preserving relative contrast between retinal layers and lesions.

8) Pathology (168K): The pathology component of MedVerse includes cytology and histopathology datasets, covering the malaria cell datasets [38, 41] and tissue microscopy dataset [35], thus comprising more than 168K images. These datasets contain high-resolution RGB microscopy images captured under diverse staining protocols, magnifications, and laboratory environments. To ensure consistent intensity scaling across heterogeneous microscopy sources, all pathology images are normalized by dividing pixel intensities by 255, producing standardized $[0, 1]$ values while preserving fine-grained cellular contrast and chromatic variations.

9) Endoscope (266K): The endoscope component of MedVerse includes gastrointestinal, urological, and surgical endoscopy datasets, covering HyperKvasir [9], Cataract-101 [61], PitVis2023 [13], Endoscapes-CVS [49] and PSI-AVA [64], thus comprising more than 266K images. It provides high-resolution RGB endoscopic frames acquired under diverse illumination settings, endoscope models, and clinical procedures. For video-based datasets, we subsample one frame every 30 frames to reduce temporal redundancy while retaining representative appearance variations across procedures. To ensure consistent intensity scaling across heterogeneous sources, all endoscopic images are normalized by dividing pixel intensities by 255, mapping them to the $[0, 1]$ range while preserving fine-grained mucosal texture characteristics.

10) Clinical photography (401K): The clinical photography component of MedVerse primarily consists of skin-lesion datasets from the ISIC archive [37] with over 401K images. It contains RGB clinical photographs captured under diverse lighting conditions, camera devices, and patient demographics, covering a wide range of benign and malignant skin lesions. To ensure consistent intensity scaling across heterogeneous sources, all clinical photographs are

normalized by dividing pixel intensities by 255, mapping them to the $[0, 1]$ range while preserving essential color cues for lesion morphology and pigmentation analysis.

B.1.2 Downstream dataset. We carefully curate our downstream datasets for a rigorous, diverse, and clinically meaningful evaluation. The selected datasets span 10 imaging modalities, 26 tasks, and a wide range of anatomical regions, enabling us to probe the capabilities of multi-modality FMs. Our choices are guided by three key principles: 1) *Clinical and anatomical coverage*: We include datasets from major diagnostic categories such as cardiac, abdominal, neurological, ophthalmic, and oncological imaging to evaluate cross-anatomy transferability. 2) *Heterogeneous modalities and acquisition settings*: By incorporating heterogeneous modalities like CT, MRI, PET, X-ray, fundus, etc., we enable the evaluations to reflect the broad variability in imaging physics, contrast mechanisms, sensor characteristics, and clinical workflows encountered in real applications. 3) *Task diversity and annotation richness*: The tasks cover organ segmentation, image-level classification, and fine-grained segmentation tasks, enabling us to assess both high-level semantic understanding and low-level anatomical precision. The specific information of the downstream task is described in Tab.2.

B.2 Implementations

Our experiments are implemented in PyTorch [54], a widely used deep learning framework. Our DEX is pretrained on two NVIDIA H100 Tensor Core GPUs with 80 GB memory, and all models are adapted on NVIDIA A100 SXUM4 GPUs with 40 GB memory. Specifically,

B.2.1 Pretraining implementation. During pretraining, our DEX is optimized by AdamW [42] with a base learning rate $r_{\text{base}} = 10^{-4}$. The initial learning rate is $r_{\text{init}} = r_{\text{base}} * B/512$, where batch size is $B = 832$. We adopt a 10-epoch warm-up schedule to stabilize training, after which the learning rate is decayed following a cosine schedule. In our adaptive frequency-aware noise mechanism, the noise standard deviation σ is initialized to 1 and annealed using the same cosine decay. The balance loss weight λ_{bal} is initialized at 0.01 and is reduced using cosine decay. For data augmentation, we apply random horizontal flipping, random affine transformations, and random resized cropping, resulting in the images with 256×256 size. Then, the images are normalized via z-score with the mean of $[0.485, 0.456, 0.406]$ and std of $[0.229, 0.224, 0.225]$ following [24]. Our DEX network adopts the ViT-B [17] as the basic architecture that applies global self-attention before our DEX module and uses the MLP layers as the experts and Director. We utilize the MAE [24] as the self-supervised learning loss $\mathcal{L}_{\text{self}}$.

B.2.2 Downstream implementation. Following [25], we utilize the fine-tuning evaluation to demonstrate the adaptation ability in downstream tasks. All segmentation tasks are implemented via the UNETR [23], and all classification tasks adopt a classification head appended to the backbone network. The gradient optimizes all parameters through the frameworks during the training. All the downstream tasks are trained by AdamW [42] optimizer with the learning rate of 10^{-4} . We adopt a 10-epoch warm-up schedule to stabilize training, after which the learning rate is decayed following a cosine schedule. For segmentation tasks, we use the sum of Dice

loss [78] and cross-entropy loss [47] between the predicted masks and ground truths to learn segmentation. For classification tasks, we utilize the cross-entropy loss [47] between the predicted categories and ground truths to train the classification.

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Table 1: The composition of our Medical Vision Universe (MedVerse) dataset. It covers 10 medical image modalities, including more than 4M images, and is able to serve the pretraining of large-scale medical vision MMFMs.

Dataset	Modality	Anatomy	Images (Slices)	Normalization	Link
AutoPET-CT [20]	CT	Whole body	558,473	$\frac{\text{clip}(x, [-200, 400]) + 200}{600}$	★
ImageCAS [74]	CT	Heart	257,496	$\frac{\text{clip}(x, [-200, 400]) + 200}{600}$	★
Topbrain-CT [71]	CT	Brain	5,647	$\frac{\text{clip}(x, [-200, 400]) + 200}{600}$	★
AutoPET-PET [20]	PET	Whole body	354,826	$\frac{\text{clip}(x, [p_{1\%}, p_{99\%}]) - \mu}{\sigma} - x_{min}$	★
AutoPET-SUV [20]	PET (SUV)	Whole body	351,900	$\frac{x_{max} - x_{min}}{\text{clip}(x, [0, 20])}$	★
Totalsegmenter-MR [2]	MR	Whole body	16,059	$\frac{\text{clip}(x, [p_{1\%}, p_{99\%}]) - x_{min}}{20}$	★
ACDC [6]	MR	Heart	1,902	$\frac{x_{max} - x_{min}}{\text{clip}(x, [p_{1\%}, p_{99\%}]) - x_{min}}$	★
MRNet [8]	MR	Knee	116,624	$\frac{x_{max} - x_{min}}{\text{clip}(x, [p_{1\%}, p_{99\%}]) - x_{min}}$	★
Duke-Breast-Cancer-MRI [60]	MR	Breast	515,747	$\frac{x_{max} - x_{min}}{\text{clip}(x, [p_{1\%}, p_{99\%}]) - x_{min}}$	★
Prostate-MRI-US-Biopsy [19, 50]	MR	Prostate	97,180	$\frac{x_{max} - x_{min}}{\text{clip}(x, [p_{1\%}, p_{99\%}]) - x_{min}}$	★
TopBrain-MR [71]	MR	Brain	4,604	$\frac{x_{max} - x_{min}}{\text{clip}(x, [p_{1\%}, p_{99\%}]) - x_{min}}$	★
BraTS-GLI [14]	MR	Brain	802,341	$\frac{x_{max} - x_{min}}{\text{clip}(x, [p_{1\%}, p_{99\%}]) - x_{min}}$	★
LLDMMR [43]	MR	Abdomen	198,918	$\frac{x_{max} - x_{min}}{\text{clip}(x, [p_{1\%}, p_{99\%}]) - x_{min}}$	★
ChestX-ray8 [68]	X-ray	Chest	112,120	$x/255$	★
CheXpert [30]	X-ray	Chest	223,648	$x/255$	★
Mammo-Bench [7]	X-ray	Breast	71,844	$x/255$	★
MURA [59]	X-ray	Bone	40,005	$x/255$	★
vzrad2 [58]	X-ray	Tooth	8,188	$x/255$	★
ADSD [12]	X-ray	Coronary artery	8,325	$x/255$	★
ARCADE [56]	X-ray	Coronary artery	3,000	$x/255$	★
XACD [45]	X-ray	Coronary artery	1,621	$x/255$	★
Thyroid Ultrasound Cine-clip [62]	Ultrasound	Thyroid	1,737	$x/255$	★
EchoNet-Dynamic [52, 53]	Ultrasound	Heart	63,867	$x/255$	★
HC18 [65]	Ultrasound	Fetal head	1,334	$x/255$	★
BUSI [3]	Ultrasound	Breast	780	$x/255$	★
UBPD [16]	Ultrasound	Brachial plexus	955	$x/255$	★
MESSIDOR [15]	Fundus	Eye	1,200	$x/255$	★
Refuge2 [18, 51]	Fundus	Eye	1,200	$x/255$	★
APTOS2019 [29]	Fundus	Eye	5,590	$x/255$	★
Eyepacs [22]	Fundus	Eye	88,702	$x/255$	★
ROSE [44]	OCT	Eye	229	$x/255$	★
OCTA500-OCTA [39]	OCT	Eye	600	$x/255$	★
OCTA500-OCT [39]	OCT	Eye	600	$x/255$	★
ZhangLabData-OCT [34]	OCT	Eye	109,309	$x/255$	★
BBBC041v1 [41]	Pathology	Blood smears	1,328	$x/255$	★
Lacuna Malaria Detection Challenge [41]	Pathology	Blood smears	3,925	$x/255$	★
Lacuna Malaria Datasets [38]	Pathology	Blood smears	5,059	$x/255$	★
SegPath [35]	Pathology	8 kinds of cells	158,687	$x/255$	★
HyperKvasir [9]	Endoscope	Stomach, intestines	110,079	$x/255$	★
Cataract-101 [61]	Endoscope	Eye	42,157	$x/255$	★
PitVis2023 [13]	Endoscope	Pituitary	96,272	$x/255$	★
Endoscapes-CVS [49]	Endoscope	Gall bladder	55,783	$x/255$	★
PSI-AVA [64]	Endoscope	Prostate	72,160	$x/255$	★
ISIC challenge 2024 training [37]	Photo	Skin	401,059	$x/255$	★
Total MedVerse			4,973,080		

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Table 2: The composition of the datasets in our downstream evaluation. It covers 26 tasks and 10 medical image modalities with varied anatomies.

Dataset	Modality	Anatomy	Task	Targets	Train	Val	Test	Link
WHS-CT (+CAT08) _S [48, 80]	CT	Heart	Segmentation	8 heart structures	5,176	0	4,096	★
KiPA22 _S [26]	CT	Kidney	Segmentation	4 kidney cancer structures	13,846	5,864	5,959	★
SISVSE _S [73]	Endoscope	Stomach	Segmentation	31 surgical scene objects	2,718	336	1,456	★
CholecSeg8k _S [28]	Endoscope	Gall bladder	Segmentation	13 cholecystectomy objects	4,320	1,120	2,640	★
IDRiD _S (task 1) [57]	Fundus	Eye	Segmentation	4 lesions and optic disc	54	0	27	★
FIVES _S [32]	Fundus	Eye	Segmentation	Fundus vessels	200	100	500	★
FeTA21 _S [55]	MR	Brain	Segmentation	13 feta brain tissues	5,120	5,120	10,240	★
CANDI _S [33]	MR	Brain	Segmentation	28 brain tissues	5,120	2,560	5,504	★
OCTDL _C [36]	OCT	Eye	Classification	7 kinds of diseases	1,030	411	623	★
ICF _S [1]	OCT	Eye	Segmentation	Cystoid Macular Edema	730	292	438	★
TNBC _S [70]	Pathology	Breast	Segmentation	Cell	20	10	20	★
LSC _C [31]	Pathology	Lymphoma	Classification	3 Lymphoma Sub-types	188	73	113	★
BCSS _S [4]	Pathology	Breast	Segmentation	5 kinds of tissues or regions	75	30	46	★
MoNuSAC _S [67]	Pathology	4 kinds of organs	Segmentation	4 kinds of nuclei	167	42	101	★
SCR _S [66]	X-ray	Chest	Segmentation	5 Chest structures	25	47	175	★★
PDCXR _C [34]	X-ray	Chest	Classification	Pneumonia	3,659	1,573	624	★
DCA1 _S [11]	X-ray	Coronary artery	Segmentation	Coronary artery	25	25	84	★
DEX _S [40]	X-ray	Brain	Segmentation	Brain artery	30	10	20	★
DSCA _S [76]	X-ray	Brain	Segmentation	Brain artery	112	44	68	★
STS-Tooth-2D _S [69]	X-ray	Tooth	Segmentation	Tooth	100	100	700	★
UNS _S [27]	Ultrasound	Nerve	Segmentation	Nerve	2,758	1,078	1,799	★
TN5000 _C [75]	Ultrasound	Thyroid nodule	Classification	Thyroid nodule	3,500	500	1,000	★
FHU _S [65]	Ultrasound	Fetus	Segmentation	Fetal head	499	200	300	★
AutoPET-orgs [21, 77]	PET	Whole body	Segmentation	12 organs or structures	18,076	6,386	10,465	★
SLSC _S [46]	Photo	Skin	Segmentation	Lesion	6,675	1,911	964	★
SLSC _C [46]	Photo	Skin	Classification	7 skin diseases	6,675	1,911	964	★

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